

Measuring the Inter-Patient Penalty in MIT-BIH Heartbeat Classification

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KNOU Statistics & Data Analysis Competition entry (Silver Prize, 2024), rebuilt from raw records in 2026 · github.com/J-Y00N/ECG-Arrhythmia-Classification-ResNet

Introduction

Beat-level evaluation lets beats from the same patient appear in both training and test sets. A widely used preprocessed release of MIT-BIH discards record identifiers, so patient-level evaluation cannot even be constructed from it. This project rebuilds the pipeline from the raw PhysioNet records, makes the evaluation protocol an experimental factor, and asks not only how large the inter-patient penalty is, but whether the usual tools can measure it.

Data

Source. MIT-BIH Arrhythmia Database via WFDB; MLII selected by name; resampled from 360 to 250 Hz; 187-sample window centred on the R peak.

Classes. Paced recordings excluded per AAMI EC57, leaving 44; AAMI classes N, S, V and F. The 15 unclassifiable beats are an observation set, never trained on.

Validation. Class counts reproduce the published DS1/DS2 distribution to within 39 of 100,733 beats.

Class–Patient Nesting

Minority classes are concentrated in a few recordings. The effective number of contributing recordings is the inverse Simpson index of a class's distribution over recordings:

$$N_{\text{eff}}(c) = 1 / \sum_r p_r^2, \quad 1 \leq N_{\text{eff}} \leq R_c$$

Class	DS1 beats	Neff	Largest recording
N normal	45,848	21.01	7%
V ventricular	3,788	7.15	26%
S supraventricular	944	4.48	41%
F fusion	414	1.24	90%

Method

Protocols. intra pools all beats and splits them 80:20; inter uses the de Chazal DS1/DS2 partition (22 recordings each), DS1 split by recording into 18 training and 4 validation.

Model. 1D residual CNN, 54,788 parameters, identical under both protocols.

Uncertainty. Cluster bootstrap over recordings (B = 2,000). Intraclass correlation ρ from a one-way random-effects model; design effect $D_{\text{eff}} = 1 + (\bar{r}\bar{n} - 1)\rho$; effective sample size n / D_{eff} .

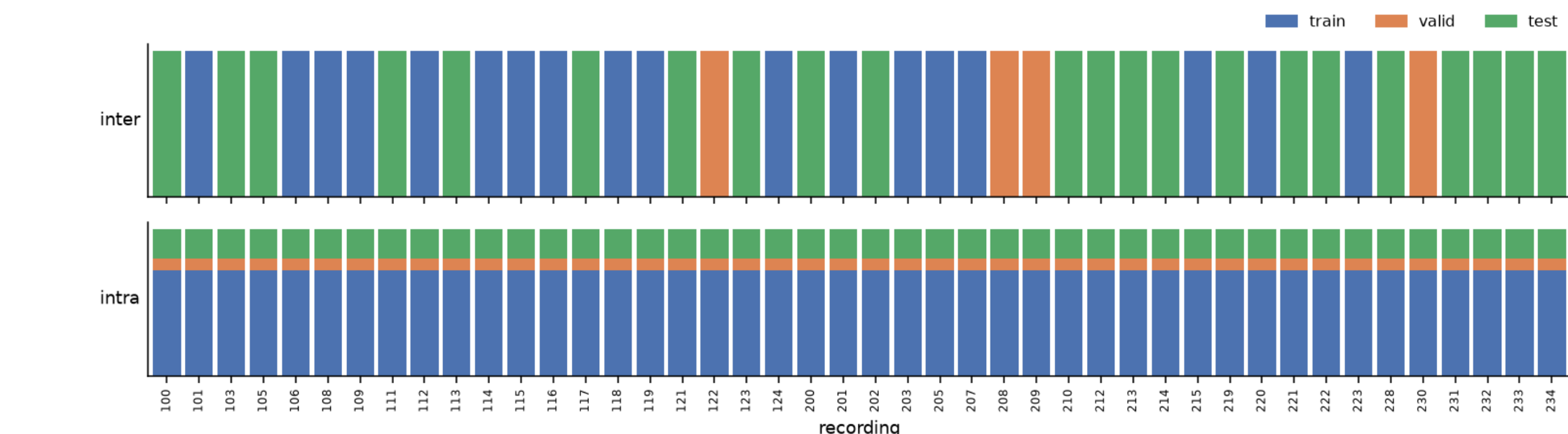


Figure 1. The 44 recordings under each protocol. Under inter a recording belongs entirely to one side; under intra every recording is split across training, validation and test.

The Protocol Gap

0.963

intra · 95% CI [0.914, 0.981]

0.592

inter · 95% CI [0.531, 0.626]

Macro F1 over N, S and V, seed 42. The cluster-bootstrap intervals do not overlap: separating patients costs 38% of the score, and the loss is concentrated in the minority classes.

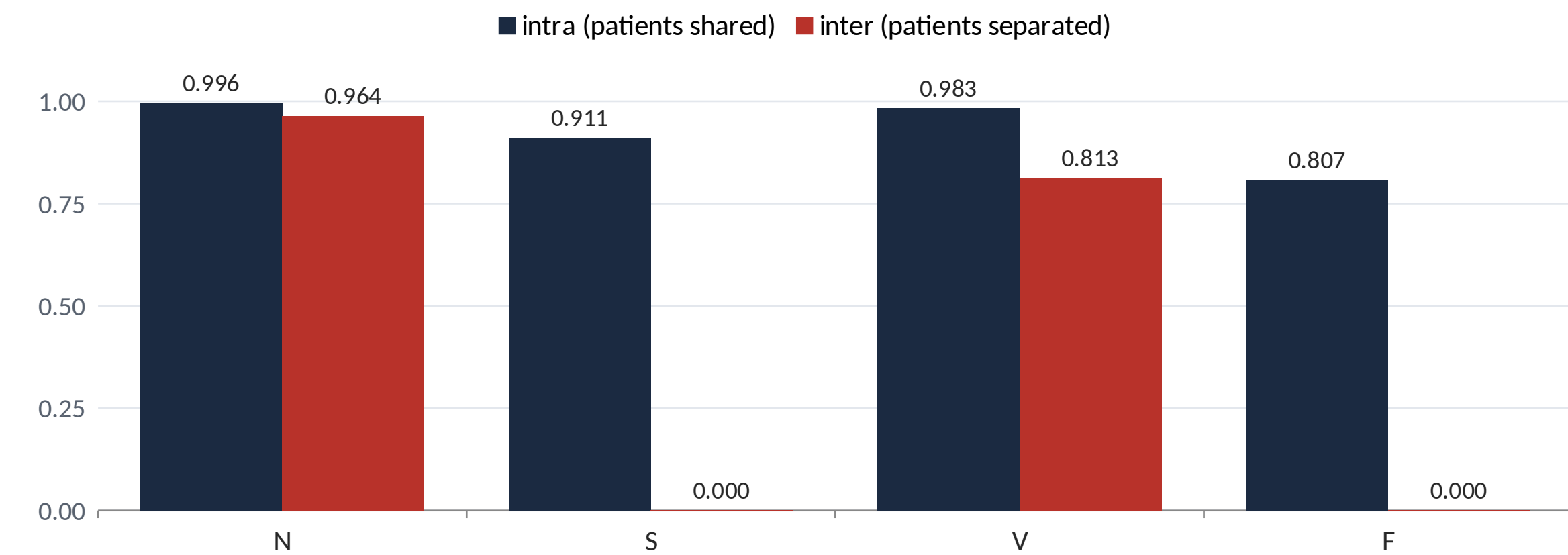


Figure 2. Per-class F1 under each protocol. Supraventricular and fusion beats fall to zero once patients are separated.

The Recording Is the Sampling Unit

0.35

intraclass correlation

795

design effect

62

effective n of 49,660 beats

In the wide-window arm, correctness is clustered within recordings: fifty thousand beats carry the information of about sixty-two independent observations. Beat-level intervals on accuracy are 24 times narrower than recording-level ones, against 28 predicted by the design effect. Under a beat-level split the same correlation reads 0.013, so that protocol hides the very structure that inflates it.

The Penalty Tracks Patient Count

Ventricular beats survive on morphology; supraventricular beats are defined by prematurity, and fusion beats come almost entirely from one patient.

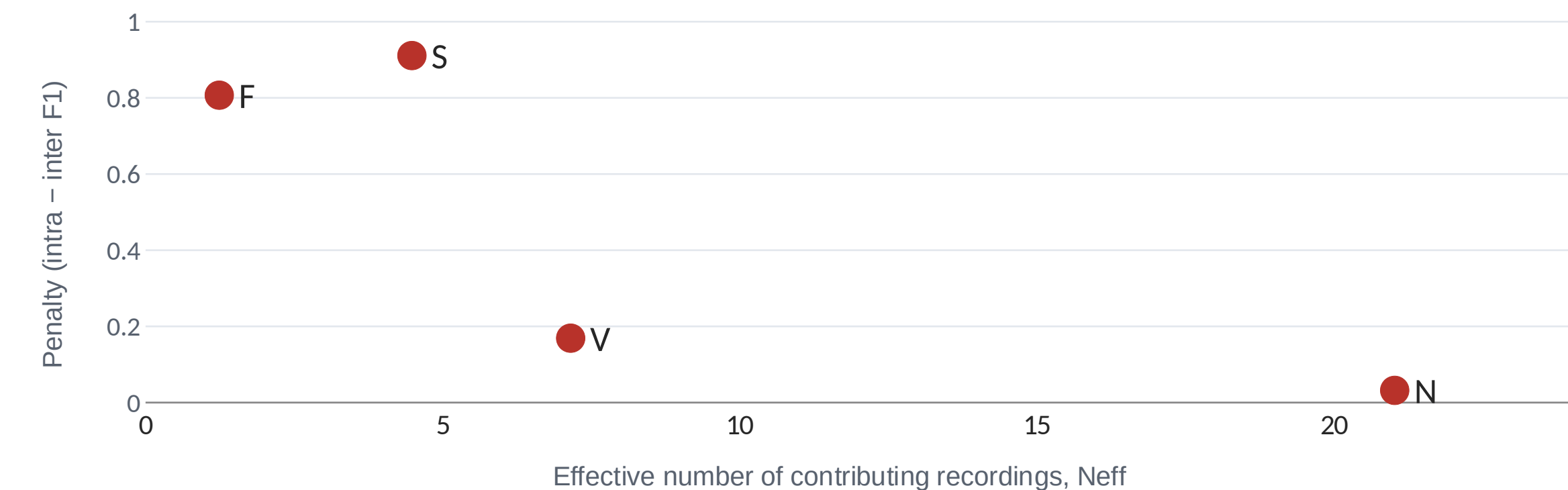
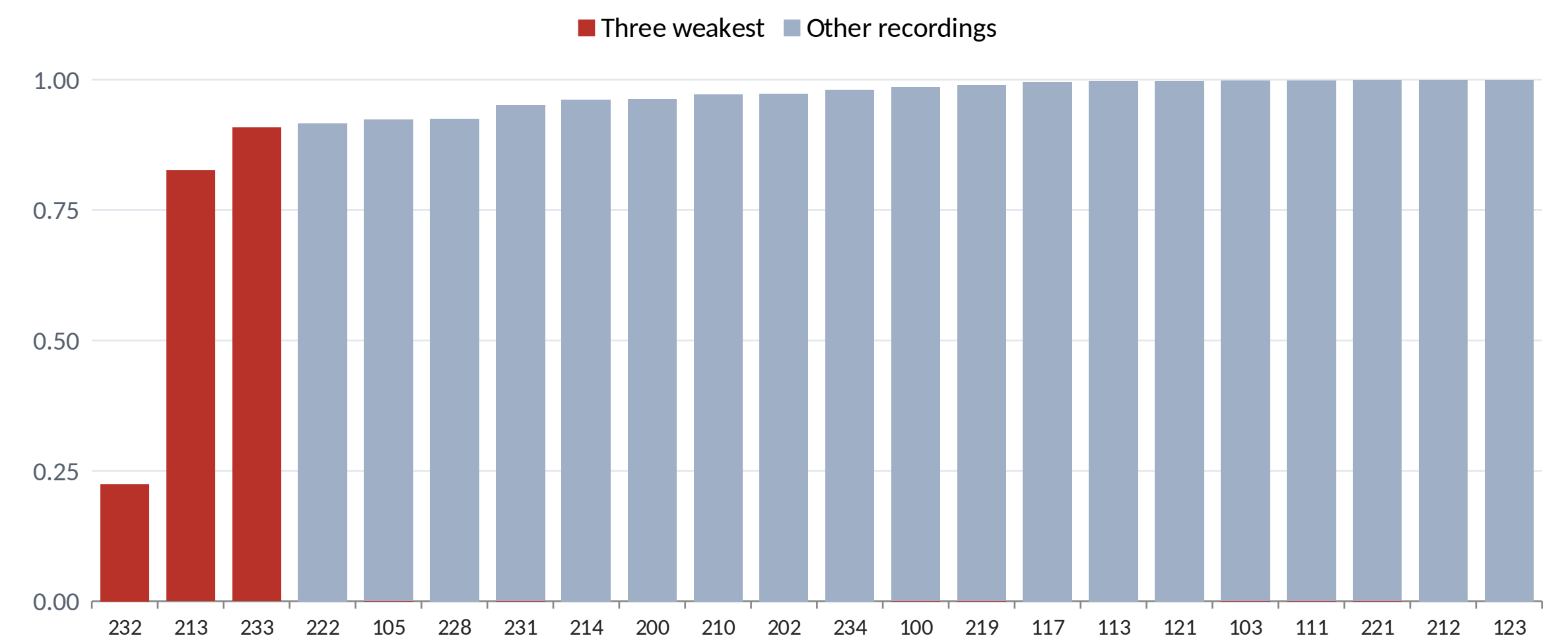


Figure 3. Per-class penalty against Neff in DS1. Spearman $\rho = -0.80$; four points support an ordering, not a p-value. Fusion has zero loss weight in this seed (record 208 held out); seeds 43–44 still score it at zero.

Where the Penalty Sits



Median accuracy 0.977; eight recordings exceed 0.99. Three recordings carry 67% of all errors. Record 232, at 0.224, is 78% supraventricular, a rhythm with no counterpart in the training half.

Figure 4. Accuracy per test recording under inter, worst first.

Calibration Hides Failure

	Confidence	Accuracy	ECE
All test beats	0.934	0.933	0.024
Record 232	0.949	0.224	0.725

Seventeen of the 22 recordings are underconfident and five overconfident; record 232 holds 3.6% of the beats, so the aggregate absorbs its failure. In automated triage, it is confidence that routes a beat to human review.

Discussion

Retrievable, not learned. A nearest-neighbour classifier reaches 0.948 under the beat-level split, against the network's 0.963.

Nothing could be shown to help. Across four loss-weight exponents, an oversampler, five representation arms and three learning rates, none improved on the simplest configuration beyond the record-level interval; two pre-stated hypotheses were refuted.

Adaptation, not generalisation. A beat-level score measures performance on patients already seen: a legitimate question, usually reported as if it were the other one.

Next. A hierarchical model with the recording as a random effect, and external validation on a third database.

References

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